

## KERALA DIPLOMA CURRICULUM REVISION 2026

Course 2073

II

As specified in the official PDF

Source-grounded

# Industrial Safety - Principles and Practices

Study the official course objectives in the source syllabus.

## OFFICIAL SOURCE DECLARATION

### How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 2073 is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

**Source file:** 2073.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

| Item                 | Official information                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Programme            | Diploma Programme                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Course code          | 2073                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Course title         | Industrial Safety - Principles and Practices                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Semester             | II                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Course category      | As specified in the official PDF                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Periods per week     | As specified in the official PDF                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Periods per semester | As specified in the official PDF                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Credits              | As specified in the official PDF                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Prerequisite         | Topic Course code Course Title Semester Engineering Physics for Applied Electrical Basic understanding of force, pressure, heat, and electricity 2002B I Technology and Computing Basic understanding of acids, bases, reactions, and 2003A Chemistry for Engineering Practices I flammable substances Basic knowledge of hazards in laboratories 2009B Engineering Workshop Practice I/II Course Outcome Blooms Taxonomy CO Course Outcome Level Explain fundamental industrial safety concepts, identify and classify hazards, and describe basic fire, CO1 Understand explosion, and toxicity principles. Explain the hierarchy of controls and describe, at an introductory level, how elimination, substitution, and CO2 Understand inherently safer approaches reduce industrial risk. Describe the purpose and basic functioning of engineering and administrative control measures such as CO3 Understand containment, detection, interlocks, permits, and emergency planning. Select appropriate fire protection and personal protective measures by relating types of hazards to CO4 Apply extinguishers and PPE and recognising their limitations. CO- PO mapping COURSE ARTICULATION MATRIX PROGRAM OUTCOMES |

| Item       | Official information                                 |
|------------|------------------------------------------------------|
| Assessment | Use the official assessment section reproduced below |

**Modules**

4 official module sections detected.

**Topic entries**

74 source-aligned topic entries retained.

**Study mode**

Theory and application emphasis.

**STUDY METHOD****How to use this lesson****First pass**

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

**Second pass**

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

**Practical evidence**

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

**Revision pass**

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

**LEARNING OUTCOMES****Objectives, outcomes and mapping****Course objectives**

- Study the official course objectives in the source syllabus.

**Course outcomes**

| CO  | Official outcome                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Study level           |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| CO1 | -- 3 -- 3 ---- 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | See official syllabus |
| CO2 | -- 3 -- 3 ---- 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | See official syllabus |
| CO3 | -- 3 -- 3 ---- 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | See official syllabus |
| CO4 | -- 3 -- 3 ---- 2 Total -- 12 -- 12 ---- 8 Correlation Level -- 3 -- 3 ---- 2<br>Diploma Curriculum Revision 2026 2073 Page #1 Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - "-" Teaching and Assessment Scheme Examination Scheme Teaching Scheme/Week Self-learning Total Credits Theory Practical (SS)/Week Hours/Semester C Total L T P S CIE ESE Total CIE ESE Total 4<br>0 0 0 6 60 4 40 60 100 0 0 0 100 L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One unit is of one-hour duration with 0.5 credit) SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination Syllabus - Major Topics Instructional Module Title Major Topics Hours L T Industrial safety introduction - safety culture - individual responsibility - safety terminology (incident - accident - near-miss - risk - unsafe-act - unsafe-condition) - Fundamentals of Industrial hazards classification - explosive limits - flammability - flash point - ignition | See official syllabus |

**CO-PO mapping evidence**

| Row | Extracted official mapping |
|-----|----------------------------|
| CO1 | CO1 -- 3 -- 3 ---- 2       |
| CO2 | CO2 -- 3 -- 3 ---- 2       |
| CO3 | CO3 -- 3 -- 3 ---- 2       |
| CO4 | CO4 -- 3 -- 3 ---- 2       |

## COVERAGE AUDIT

# Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

### Module coverage

| Module   | Official heading                                                                                    | Topic entries | Hours        |
|----------|-----------------------------------------------------------------------------------------------------|---------------|--------------|
| Module 1 | 15 0                                                                                                | 5             | As specified |
| Module 2 | Inherently Safer safety benefits - inventory reduction - safer process routes - advantages and 15 0 | 21            | As specified |
| Module 3 | Define engineering control and explain its purpose.                                                 | 24            | As specified |
| Module 4 | Fire extinguishing Explain fire-extinguishing techniques - cooling,                                 | 24            | As specified |

**MODULE 1****15 0**

This module is presented from the official syllabus scope for Industrial Safety - Principles and Practices. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**  
official hours

**CO-linked**  
see outcomes

**Learning goals**

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

**Key facts**

- Official module title: 15 0
- Official duration: As specified
- Official topic entries captured: 5
- The full source excerpt is preserved below for coverage checking.

## ▼ Industrial Safety temperature - toxicity - routes of entry - threshold exposure limits - chemistry of fire -

Industrial Safety temperature - toxicity - routes of entry - threshold exposure limits - chemistry of fire - is an official topic in Module 1 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Industrial Safety temperature - toxicity - routes of entry - threshold exposure limits - chemistry of fire - using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, industrial safety temperature - toxicity - routes of entry - threshold exposure limits - chemistry of fire - should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide



| Aspect              | What to prepare                                                                                                                                        |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Industrial Safety temperature - toxicity - routes of entry - threshold exposure limits - chemistry of fire - means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                                                |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                                      |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-Industrial Safety temperature - toxicity - routes of entry - threshold exposure limits - chemistry of fire -  
 പദപരിചയം definition/meaning പദപരിചയം. പദപരിചയം പദപരിചയം  
 പദപരിചയം, steps, application പദപരിചയം പദപരിചയം പദപരിചയം. Technical terms  
 English-പദപരിചയം പദപരിചയം.

## ▼ fire triangle - fire tetrahedron - classification of fires - causes of fire - major industrial

fire triangle - fire tetrahedron - classification of fires - causes of fire - major industrial is an official topic in Module 1 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify fire triangle - fire tetrahedron - classification of fires - causes of fire - major industrial using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, fire triangle - fire tetrahedron - classification of fires - causes of fire - major industrial should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect | What to prepare |
|--------|-----------------|
|--------|-----------------|

| Aspect              | What to prepare                                                                                                                          |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What fire triangle – fire tetrahedron – classification of fires – causes of fire – major industrial means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                                  |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                        |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

[illegible]

## ▼ accidents – safety lessons.

accidents – safety lessons. is an official topic in Module 1 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify accidents – safety lessons. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, accidents – safety lessons. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                       |
|---------------------|-----------------------------------------------------------------------|
| Definition/identity | What accidents – safety lessons. means in the official course context |
| Study lens          | Principle → components or stages → result → application               |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-accidents – safety lessons.

അപകടങ്ങളുടെ നിർവ്വചനം/അർത്ഥം, കാരണങ്ങൾ, തടയൽ നടപടികൾ, steps, application എന്നിവയെക്കുറിച്ചുള്ള വിവരങ്ങൾ. Technical terms English-ൽ എഴുതുക.

## ▼ Hierarchy of controls - significance in risk reduction - source-level control - inherently

Hierarchy of controls - significance in risk reduction - source-level control - inherently is an official topic in Module 1 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Hierarchy of controls - significance in risk reduction - source-level control - inherently using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, hierarchy of controls - significance in risk reduction - source-level control - inherently should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect | What to prepare |
|--------|-----------------|
|--------|-----------------|

| Aspect              | What to prepare                                                                                                                      |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Hierarchy of controls – significance in risk reduction – source-level control – inherently means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                              |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                    |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-□□ Hierarchy of controls – significance in risk reduction – source-level control – inherently □□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□. □□□□□ □□□□□ □□□□□□, steps, application □□□□□ □□□□□□□□ □□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□.

## ▼ Hazard Reduction and safer design philosophy - substitution of hazardous materials - environmental and

Hazard Reduction and safer design philosophy - substitution of hazardous materials - environmental and is an official topic in Module 1 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Hazard Reduction and safer design philosophy - substitution of hazardous materials - environmental and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, hazard reduction and safer design philosophy - substitution of hazardous materials - environmental and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide





**MODULE 2**

# Inherently Safer safety benefits - inventory reduction - safer process routes - advantages and 15 0

This module is presented from the official syllabus scope for Industrial Safety - Principles and Practices. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**  
official hours

**CO-linked**  
see outcomes

## Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

## Key facts

- Official module title: Inherently Safer safety benefits - inventory reduction - safer process routes - advantages and 15 0
- Official duration: As specified
- Official topic entries captured: 21
- The full source excerpt is preserved below for coverage checking.

## ▼ Inherently Safer safety benefits – inventory reduction – safer process routes – advantages and 15 0

Inherently Safer safety benefits – inventory reduction – safer process routes – advantages and 15 0 is an official topic in Module 2 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Inherently Safer safety benefits – inventory reduction – safer process routes – advantages and 15 0 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, inherently safer safety benefits – inventory reduction – safer process routes – advantages and 15 0 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                                                                               |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Inherently Safer safety benefits – inventory reduction – safer process routes – advantages and 15 0 means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                                       |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                             |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-□□ Inherently Safer safety benefits – inventory reduction – safer process routes – advantages and 15 0 □□□□□□□□□□ □□□□ definition/meaning □□□□□□□□□□. □□□□□ □□□□□ □□□□□□, steps, application □□□□□ □□□□□□□□ □□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□.

## ▼ Approaches limitations of source-level control - awareness of hazard analysis methods (PHA -

Approaches limitations of source-level control - awareness of hazard analysis methods (PHA - is an official topic in Module 2 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Approaches limitations of source-level control - awareness of hazard analysis methods (PHA - using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, approaches limitations of source-level control - awareness of hazard analysis methods (pha - should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect | What to prepare |
|--------|-----------------|
|--------|-----------------|

| Aspect              | What to prepare                                                                                                                        |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Approaches limitations of source-level control – awareness of hazard analysis methods (PHA – means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                                |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                      |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-എന്നിരിക്കുന്നതാണ് Approaches limitations of source-level control – awareness of hazard analysis methods (PHA – ഏതെങ്കിലും ഏതെങ്കിലും definition/meaning എഴുതേണ്ടതാണ്. ഏതെങ്കിലും ഏതെങ്കിലും ഏതെങ്കിലും, steps, application എഴുതേണ്ടതാണ് ഏതെങ്കിലും ഏതെങ്കിലും. Technical terms English-ൽ എഴുതേണ്ടതാണ്.

## ▼ What-If - JSA - HAZOP - FTA) - purpose and basic application of each method.

What-If - JSA - HAZOP - FTA) - purpose and basic application of each method. is an official topic in Module 2 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify What-If - JSA - HAZOP - FTA) - purpose and basic application of each method. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, what-if - jsa - hazop - fta) - purpose and basic application of each method. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect | What to prepare |
|--------|-----------------|
|--------|-----------------|

| Aspect              | What to prepare                                                                                                        |
|---------------------|------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What What-If – JSA – HAZOP – FTA) – purpose and basic application of each method. means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                      |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-എന്തെങ്കിലും JSA – HAZOP – FTA) – purpose and basic application of each method. ഏതെങ്കിലും ഏതെങ്കിലും definition/meaning എന്തെങ്കിലും. എന്തെങ്കിലും എന്തെങ്കിലും steps, application എന്തെങ്കിലും എന്തെങ്കിലും. Technical terms English-എന്തെങ്കിലും എന്തെങ്കിലും.



## ▼ Engineering controls - limitations of engineering controls - process containment -

Engineering controls - limitations of engineering controls - process containment - is an official topic in Module 2 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Engineering controls - limitations of engineering controls - process containment - using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, engineering controls - limitations of engineering controls - process containment - should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect | What to prepare |
|--------|-----------------|
|--------|-----------------|

| Aspect              | What to prepare                                                                                                              |
|---------------------|------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Engineering controls – limitations of engineering controls – process containment – means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                      |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                            |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-എംഗിനീയറിംഗ് കൺട്രോളുകൾ - എൻജിനീയറിംഗ് കൺട്രോളുകൾ - പ്രോസസ്സ് കൺടൈൻമെന്റ് - നിർവ്വചനം/അർത്ഥം നിർവ്വചിക്കുക. നിർവ്വചനം നിർവ്വചിക്കുക, steps, application നിർവ്വചിക്കുക. Technical terms English-ൽ നിർവ്വചിക്കുക.

## ▼ enclosures - barriers - segregation - ventilation - preventing unsafe pressure or

enclosures - barriers - segregation - ventilation - preventing unsafe pressure or is an official topic in Module 2 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify enclosures - barriers - segregation - ventilation - preventing unsafe pressure or using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, enclosures - barriers - segregation - ventilation - preventing unsafe pressure or should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect | What to prepare |
|--------|-----------------|
|--------|-----------------|

| Aspect              | What to prepare                                                                                                             |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What enclosures – barriers – segregation – ventilation – preventing unsafe pressure or means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                     |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                           |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-എൻക്ലോച്ചറുകൾ - ബാരിയറുകൾ - സെഗ്രഗേഷൻ - വെന്റിലേഷൻ - പ്രിവെന്റിംഗ് അസേഫ് പ്രഷ്യർ അല്ലെങ്കിൽ മറ്റ് ഓഫീഷ്യൽ കോഴ്സ് കോണ്ടക്സ്റ്റ് ഓഫ് ഡിഫിനിഷൻ/മീനിംഗ് എക്സ്പ്ലൈൻ ചെയ്യുക. എക്സ്പ്ലൈൻ ചെയ്യുക ഓഫീഷ്യൽ കോണ്ടക്സ്റ്റ്, steps, application എക്സ്പ്ലൈൻ ചെയ്യുക ഓഫീഷ്യൽ കോണ്ടക്സ്റ്റ്. Technical terms English-ൽ എക്സ്പ്ലൈൻ ചെയ്യുക.

## ▼ Engineering and

Engineering and is an official topic in Module 2 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Engineering and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, engineering and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                           |
|---------------------|-----------------------------------------------------------|
| Definition/identity | What Engineering and means in the official course context |
| Study lens          | Principle → components or stages → result → application   |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-എംഗിനീയറിംഗ് ആന്റ് സുരക്ഷാ മാനദണ്ഡങ്ങൾ എന്നതിൽ നിന്ന് definition/meaning എടുത്ത് എഴുതുക. ഉദാഹരണം: സുരക്ഷാ മാനദണ്ഡങ്ങൾ, steps, application എന്നിവയ്ക്ക് ഉദാഹരണം നൽകുക. Technical terms English-ൽ എഴുതുക.

## ▼ Administrative Control 15 0

Administrative Control 15 0 is an official topic in Module 2 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Administrative Control 15 0 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, administrative control 15 0 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                       |
|---------------------|-----------------------------------------------------------------------|
| Definition/identity | What Administrative Control 15 0 means in the official course context |
| Study lens          | Principle → components or stages → result → application               |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-എന്നിവിടെ Administrative Control 15 0

എന്നിവിടെ definition/meaning എന്നിവിടെ. എന്നിവിടെ എന്നിവിടെ എന്നിവിടെ, steps, application എന്നിവിടെ എന്നിവിടെ എന്നിവിടെ. Technical terms English-എന്നിവിടെ എന്നിവിടെ എന്നിവിടെ.



## ▼ III systems - automatic protection - administrative controls - limitations of administrative

III systems - automatic protection - administrative controls - limitations of administrative is an official topic in Module 2 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify III systems - automatic protection - administrative controls - limitations of administrative using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, iii systems - automatic protection - administrative controls - limitations of administrative should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect | What to prepare |
|--------|-----------------|
|--------|-----------------|

| Aspect              | What to prepare                                                                                                                        |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What III systems – automatic protection – administrative controls – limitations of administrative means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                                |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                      |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-III systems – automatic protection – administrative controls – limitations of administrative definition/meaning. ഏകദേശം, steps, application. Technical terms English-III.

## ▼ Measures

Measures is an official topic in Module 2 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Measures using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, measures should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                         |
|---------------------|---------------------------------------------------------|
| Definition/identity | What Measures means in the official course context      |
| Study lens          | Principle → components or stages → result → application |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-ഇരുപതു അളവുകൾ (Measures) എന്നതിന്റെ നിർവ്വചനം/അർത്ഥം (definition/meaning) എഴുതുക. അളവുകൾ എങ്ങനെ ഉപയോഗിക്കുന്നു, steps, application എന്നിവയെക്കുറിച്ചും എഴുതുക. Technical terms English-ൽ എഴുതുക.

## ▼ measures – permit-to-work systems – lockout/tagout intent – onsite and offsite

measures – permit-to-work systems – lockout/tagout intent – onsite and offsite is an official topic in Module 2 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify measures – permit-to-work systems – lockout/tagout intent – onsite and offsite using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, measures – permit-to-work systems – lockout/tagout intent – onsite and offsite should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect | What to prepare |
|--------|-----------------|
|--------|-----------------|

| Aspect              | What to prepare                                                                                                          |
|---------------------|--------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What measures – permit-to-work systems – lockout/tagout intent – onsite and offsite means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                  |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                        |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-എന്ന measures – permit-to-work systems – lockout/tagout intent – onsite and offsite എന്നിവയുടെ definition/meaning എന്തെന്നും. എന്നിവയുടെ components, steps, application എന്നിവയെക്കുറിച്ചും. Technical terms English-ൽ എഴുതുക.

## ▼ emergency planning.

emergency planning. is an official topic in Module 2 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify emergency planning. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, emergency planning. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                               |
|---------------------|---------------------------------------------------------------|
| Definition/identity | What emergency planning. means in the official course context |
| Study lens          | Principle → components or stages → result → application       |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-എmergency planning. എdefinition/meaning എ. എ steps, application എ. Technical terms English-എ



## ▼ Fire extinguishment - classification of fires - selection of suitable extinguishing media

Fire extinguishment – classification of fires – selection of suitable extinguishing media is an official topic in Module 2 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Fire extinguishment – classification of fires – selection of suitable extinguishing media using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, fire extinguishment – classification of fires – selection of suitable extinguishing media should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect | What to prepare |
|--------|-----------------|
|--------|-----------------|

| Aspect              | What to prepare                                                                                                                     |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Fire extinguishment – classification of fires – selection of suitable extinguishing media means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                             |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                   |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-☐ Fire extinguishment – classification of fires – selection of suitable extinguishing media ☐☐☐☐☐☐☐☐☐ definition/meaning ☐☐☐☐☐☐☐☐☐. ☐☐☐☐☐ ☐☐☐☐☐ ☐☐☐☐☐☐☐, steps, application ☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐. Technical terms English-☐ ☐☐☐☐☐☐☐☐☐☐☐.

## ▼ Fire Protection and

Fire Protection and is an official topic in Module 2 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Fire Protection and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, fire protection and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                               |
|---------------------|---------------------------------------------------------------|
| Definition/identity | What Fire Protection and means in the official course context |
| Study lens          | Principle → components or stages → result → application       |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-പ്രതിരോധം Fire Protection and പ്രതിരോധം പ്രതിരോധം definition/meaning പ്രതിരോധം പ്രതിരോധം. പ്രതിരോധം പ്രതിരോധം പ്രതിരോധം, steps, application പ്രതിരോധം പ്രതിരോധം പ്രതിരോധം. Technical terms English-പ്രതിരോധം പ്രതിരോധം.

## ▼ Personal Protective 15 0

Personal Protective 15 0 is an official topic in Module 2 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Personal Protective 15 0 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, personal protective 15 0 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                    |
|---------------------|--------------------------------------------------------------------|
| Definition/identity | What Personal Protective 15 0 means in the official course context |
| Study lens          | Principle → components or stages → result → application            |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-പ്രതിരോധം Personal Protective 15 0 പ്രതിരോധം പ്രതിരോധം definition/meaning പ്രതിരോധം പ്രതിരോധം. പ്രതിരോധം പ്രതിരോധം പ്രതിരോധം, steps, application പ്രതിരോധം പ്രതിരോധം പ്രതിരോധം. Technical terms English-പ്രതിരോധം പ്രതിരോധം.

## ▼ IV extinguishers - PPE - PPE limitations - classification of PPE - PPE selection and use -

IV extinguishers - PPE - PPE limitations - classification of PPE - PPE selection and use - is an official topic in Module 2 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify IV extinguishers - PPE - PPE limitations - classification of PPE - PPE selection and use - using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, iv extinguishers - ppe - ppe limitations - classification of ppe - ppe selection and use - should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect | What to prepare |
|--------|-----------------|
|--------|-----------------|

| Aspect              | What to prepare                                                                                                                      |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What IV extinguishers – PPE – PPE limitations – classification of PPE – PPE selection and use – means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                              |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                    |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-ഇവ IV extinguishers – PPE – PPE limitations – classification of PPE – PPE selection and use – പദപരിചയം പദപരിചയം definition/meaning പദപരിചയം പദപരിചയം. പദപരിചയം പദപരിചയം പദപരിചയം, steps, application പദപരിചയം പദപരിചയം പദപരിചയം. Technical terms English-ഇ പദപരിചയം പദപരിചയം പദപരിചയം.



## ▼ Equipment

Equipment is an official topic in Module 2 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Equipment using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, equipment should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                         |
|---------------------|---------------------------------------------------------|
| Definition/identity | What Equipment means in the official course context     |
| Study lens          | Principle → components or stages → result → application |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-എക്സ്ക്വൈപ്പ്മെന്റ് നിർവ്വചനം/അർത്ഥം എഴുതുക. എക്സ്ക്വൈപ്പ്മെന്റ് ഉപയോഗം, steps, application എഴുതുക. Technical terms English-ൽ എഴുതുക.

## ▼ awareness of modern wearable safety technologies.

awareness of modern wearable safety technologies. is an official topic in Module 2 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify awareness of modern wearable safety technologies. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, awareness of modern wearable safety technologies. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                             |
|---------------------|---------------------------------------------------------------------------------------------|
| Definition/identity | What awareness of modern wearable safety technologies. means in the official course context |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Study lens    | Principle → components or stages → result → application                           |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-ഈ അറിവ് ആധുനിക ഓരോരുത്തരുടെയും സുരക്ഷാ സാങ്കേതികവിദ്യകളെക്കുറിച്ചാണ്. ഈ വിഷയത്തിന് നിർവ്വചനം/അർത്ഥം നിർവ്വചിക്കുക. ഈ വിഷയത്തിന് ഘട്ടങ്ങൾ, പ്രയോഗം എന്നിവയെക്കുറിച്ച് വിശദീകരിക്കുക. Technical terms English-ഈ വിഷയത്തിന് ഏതെങ്കിലും സാങ്കേതിക പദങ്ങൾ ഉണ്ടാകാം.

## ▼ Detailed Syllabus

Detailed Syllabus is an official topic in Module 2 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Detailed Syllabus using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, detailed syllabus should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                             |
|---------------------|-------------------------------------------------------------|
| Definition/identity | What Detailed Syllabus means in the official course context |
| Study lens          | Principle → components or stages → result → application     |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-എന്ന Detailed Syllabus പരീക്ഷാസൂചികയിൽ നിന്ന് definition/meaning പരീക്ഷാസൂചികയിൽ. പരീക്ഷാ സൂചികയിൽ പരീക്ഷാസൂചിക, steps, application പരീക്ഷാ സൂചികയിൽ പരീക്ഷാസൂചിക. Technical terms English-ൽ പരീക്ഷാ സൂചികയിൽ പരീക്ഷാസൂചിക.

## ▼ Mode of

Mode of is an official topic in Module 2 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Mode of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, mode of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                         |
|---------------------|---------------------------------------------------------|
| Definition/identity | What Mode of means in the official course context       |
| Study lens          | Principle → components or stages → result → application |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-എന്നിരിക്കുന്നതാണ് നിലവിലുള്ള നിർവ്വചനം/അർത്ഥം. നിർവ്വചനം, അർത്ഥം, ഘട്ടങ്ങൾ, steps, application എന്നിവയെക്കുറിച്ചുള്ള വിവരങ്ങൾ. Technical terms English-ൽ നിന്നും മലയാളത്തിലേക്ക്.



## ▼ Mapped Instructional

Mapped Instructional is an official topic in Module 2 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Mapped Instructional using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, mapped instructional should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                |
|---------------------|----------------------------------------------------------------|
| Definition/identity | What Mapped Instructional means in the official course context |
| Study lens          | Principle → components or stages → result → application        |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-എന്നിവിടെ Mapped Instructional ഉള്ളതിനെക്കുറിച്ച് definition/meaning ഉള്ളതുകൊണ്ട്. ഉള്ളതിനെക്കുറിച്ച് ഉള്ളതിനെക്കുറിച്ച്, steps, application ഉള്ളതിനെക്കുറിച്ച് ഉള്ളതിനെക്കുറിച്ച്. Technical terms English-ൽ ഉള്ളതിനെക്കുറിച്ച് ഉള്ളതിനെക്കുറിച്ച്.

## ▼ Subtopic Student Learning Outcome delivery Taxonomy Level

Subtopic Student Learning Outcome delivery Taxonomy Level is an official topic in Module 2 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Subtopic Student Learning Outcome delivery Taxonomy Level using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, subtopic student learning outcome delivery taxonomy level should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                                     |
|---------------------|-----------------------------------------------------------------------------------------------------|
| Definition/identity | What Subtopic Student Learning Outcome delivery Taxonomy Level means in the official course context |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Study lens    | Principle → components or stages → result → application                           |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-Subtopic Student Learning Outcome delivery Taxonomy Level definition/meaning. Steps, application. Technical terms English-.

► **Official syllabus detail and terminology (expand in digital version)**

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

**MODULE 3**

# Define engineering control and explain its purpose.

This module is presented from the official syllabus scope for Industrial Safety - Principles and Practices. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**  
official hours

**CO-linked**  
see outcomes

## Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

## Key facts

- Official module title: Define engineering control and explain its purpose.
- Official duration: As specified
- Official topic entries captured: 24
- The full source excerpt is preserved below for coverage checking.

## ▼ Define engineering control and explain its purpose.

Define engineering control and explain its purpose. is an official topic in Module 3 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Define engineering control and explain its purpose. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, define engineering control and explain its purpose. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                               |
|---------------------|-----------------------------------------------------------------------------------------------|
| Definition/identity | What Define engineering control and explain its purpose. means in the official course context |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Study lens    | Principle → components or stages → result → application                           |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3- Define engineering control and explain its purpose. definition/meaning. steps, application. Technical terms English-.

## ▼ Introduction to Differentiate between source control and engineering

Introduction to Differentiate between source control and engineering is an official topic in Module 3 of Industrial Safety - Principles and Practices.

Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Introduction to Differentiate between source control and engineering using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, introduction to differentiate between source control and engineering should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect | What to prepare |
|--------|-----------------|
|--------|-----------------|



| Aspect              | What to prepare                                                                                                |
|---------------------|----------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Introduction to Differentiate between source control and engineering means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                        |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                              |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-എന്നു തുടങ്ങുന്ന Introduction to Differentiate between source control and engineering എന്നതിന്റെ definition/meaning എഴുതുക. തുടർന്ന് ഈ വിഷയം, steps, application എന്നിവയെക്കുറിച്ചും എഴുതുക. Technical terms English-ൽ എഴുതുക.

## ▼ Engineering Controls control.

Engineering Controls control. is an official topic in Module 3 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Engineering Controls control. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, engineering controls control. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                         |
|---------------------|-------------------------------------------------------------------------|
| Definition/identity | What Engineering Controls control. means in the official course context |
| Study lens          | Principle → components or stages → result → application                 |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-എംഗിനീയറിംഗ് കൺട്രോൾസ് കൺട്രോൾ.

എംഗിനീയറിംഗ് കൺട്രോൾസ് definition/meaning എംഗിനീയറിംഗ് കൺട്രോൾസ്. എംഗിനീയറിംഗ് കൺട്രോൾസ്, steps, application എംഗിനീയറിംഗ് കൺട്രോൾസ്. Technical terms English-ൽ എംഗിനീയറിംഗ് കൺട്രോൾസ്.

## ▼ Identify the limitations of engineering control.

Identify the limitations of engineering control. is an official topic in Module 3 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Identify the limitations of engineering control. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, identify the limitations of engineering control. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                            |
|---------------------|--------------------------------------------------------------------------------------------|
| Definition/identity | What Identify the limitations of engineering control. means in the official course context |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Study lens    | Principle → components or stages → result → application                           |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-ഓ ഓ ഓ Identify the limitations of engineering control. ഓ ഓ ഓ definition/meaning ഓ ഓ ഓ. ഓ ഓ ഓ ഓ ഓ, steps, application ഓ ഓ ഓ ഓ ഓ. Technical terms English-ഓ ഓ ഓ ഓ ഓ.

## ▼ Summarise process containment and isolation.

Summarise process containment and isolation. is an official topic in Module 3 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Summarise process containment and isolation. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, summarise process containment and isolation. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                        |
|---------------------|----------------------------------------------------------------------------------------|
| Definition/identity | What Summarise process containment and isolation. means in the official course context |



## ▼ Process containment

Process containment is an official topic in Module 3 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Process containment using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, process containment should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                               |
|---------------------|---------------------------------------------------------------|
| Definition/identity | What Process containment means in the official course context |
| Study lens          | Principle → components or stages → result → application       |



| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-പ്രക്രിയ അടയ്ക്കൽ പ്രക്രിയയുടെ definition/meaning, steps, application, Technical terms English-ൽ വിശദീകരിക്കുക.

## ▼ Describe at an introductory level how containment and L CO3 Understand 2

Describe at an introductory level how containment and L CO3 Understand 2 is an official topic in Module 3 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Describe at an introductory level how containment and L CO3 Understand 2 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, describe at an introductory level how containment and L co3 understand 2 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect | What to prepare |
|--------|-----------------|
|--------|-----------------|

**Malayalam support:** Module 3-Describe at an introductory level how containment and L CO3 Understand 2 ഓരോന്നിനും നിർവ്വചനം/അർത്ഥം, ഓരോന്നിന്റെയും പ്രയോഗങ്ങൾ, steps, application എന്നിവ ഉൾപ്പെടെ വിശദീകരിക്കുക. Technical terms English-ൽ പരിഭാഷിക്കുക.

## ▼ and isolation

and isolation is an official topic in Module 3 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify and isolation using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, and isolation should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                         |
|---------------------|---------------------------------------------------------|
| Definition/identity | What and isolation means in the official course context |
| Study lens          | Principle → components or stages → result → application |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-പ്രതിരോധം and isolation പ്രതിരോധം definition/meaning പ്രതിരോധം. പ്രതിരോധം പ്രതിരോധം പ്രതിരോധം, steps, application പ്രതിരോധം പ്രതിരോധം പ്രതിരോധം. Technical terms English-ൽ പ്രതിരോധം പ്രതിരോധം.

## ▼ physical barriers prevent release or exposure.

physical barriers prevent release or exposure. is an official topic in Module 3 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify physical barriers prevent release or exposure. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, physical barriers prevent release or exposure. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                          |
|---------------------|------------------------------------------------------------------------------------------|
| Definition/identity | What physical barriers prevent release or exposure. means in the official course context |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Study lens    | Principle → components or stages → result → application                           |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3- physical barriers prevent release or exposure. definition/meaning . steps, application . Technical terms English- .

## ▼ Outline the ventilation and exposure control.

Outline the ventilation and exposure control. is an official topic in Module 3 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Outline the ventilation and exposure control. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, outline the ventilation and exposure control. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                         |
|---------------------|-----------------------------------------------------------------------------------------|
| Definition/identity | What Outline the ventilation and exposure control. means in the official course context |



| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Study lens    | Principle → components or stages → result → application                           |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-ഓൺ ഔട്ട്ലൈൻ വെന്റിലേഷൻ ആൻഡ് എക്സപ്പോഷർ കൺട്രോൾ. ഓരോ ഓരോ ഓരോ definition/meaning ഓരോ ഓരോ. ഓരോ ഓരോ ഓരോ, steps, application ഓരോ ഓരോ ഓരോ. Technical terms English-ഓ ഓരോ ഓരോ.

## ▼ Ventilation and Exposure

Ventilation and Exposure is an official topic in Module 3 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Ventilation and Exposure using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, ventilation and exposure should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                    |
|---------------------|--------------------------------------------------------------------|
| Definition/identity | What Ventilation and Exposure means in the official course context |
| Study lens          | Principle → components or stages → result → application            |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-വентиляция and Exposure വെള്ളം വെള്ളം വെള്ളം വെള്ളം വെള്ളം definition/meaning വെള്ളം വെള്ളം വെള്ളം വെള്ളം വെള്ളം. വെള്ളം വെള്ളം വെള്ളം വെള്ളം വെള്ളം, steps, application വെള്ളം വെള്ളം വെള്ളം വെള്ളം വെള്ളം. Technical terms English-വെള്ളം വെള്ളം വെള്ളം വെള്ളം വെള്ളം.

## ▼ State the role of ventilation in reducing employee L CO3 Understand 1

State the role of ventilation in reducing employee L CO3 Understand 1 is an official topic in Module 3 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify State the role of ventilation in reducing employee L CO3 Understand 1 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, state the role of ventilation in reducing employee L co3 understand 1 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect | What to prepare |
|--------|-----------------|
|--------|-----------------|

| Aspect              | What to prepare                                                                                                    |
|---------------------|--------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What State the role of ventilation in reducing employee L CO3<br>Understand 1 means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                            |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                  |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3- State the role of ventilation in reducing employee L CO3 Understand 1 definition/meaning . steps, application . Technical terms English- .

## ▼ exposure.

exposure. is an official topic in Module 3 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify exposure. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, exposure. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                         |
|---------------------|---------------------------------------------------------|
| Definition/identity | What exposure. means in the official course context     |
| Study lens          | Principle → components or stages → result → application |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-എക്സപ്പോഷറും. ഏകദേശം നിർവ്വചനം/അർത്ഥം എഴുതുക. എഴുതുക എഴുതുക എഴുതുക, steps, application എഴുതുക എഴുതുക എഴുതുക. Technical terms English-ൽ എഴുതുക എഴുതുക എഴുതുക.

## ▼ Pressure and Outline pressure and temperature control systems.

Pressure and Outline pressure and temperature control systems. is an official topic in Module 3 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Pressure and Outline pressure and temperature control systems. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, pressure and outline pressure and temperature control systems. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect | What to prepare |
|--------|-----------------|
|--------|-----------------|



| Aspect              | What to prepare                                                                                          |
|---------------------|----------------------------------------------------------------------------------------------------------|
| Definition/identity | What Pressure and Outline pressure and temperature control systems. means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                  |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                        |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-പ്രഷർ ആന്റ് ഔട്ട്ലൈൻ പ്രഷർ ആന്റ് ടെമ്പറേച്ചർ കൺട്രോൾ സിസ്റ്റം. പദപരിചയം definition/meaning പദപരിചയം. പദപരിചയം പദപരിചയം, steps, application പദപരിചയം പദപരിചയം പദപരിചയം. Technical terms English-ൽ പദപരിചയം പദപരിചയം.

## ▼ Temperature Control Explain in simple terms why systems are provided to L CO3 Understand 1

Temperature Control Explain in simple terms why systems are provided to L CO3 Understand 1 is an official topic in Module 3 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Temperature Control Explain in simple terms why systems are provided to L CO3 Understand 1 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, temperature control explain in simple terms why systems are provided to L co3 understand 1 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect | What to prepare |
|--------|-----------------|
|--------|-----------------|

| Aspect              | What to prepare                                                                                                                      |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Temperature Control Explain in simple terms why systems are provided to L CO3 Understand 1 means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                              |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                    |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-താപനിയന്ത്രണം Temperature Control Explain in simple terms why systems are provided to L CO3 Understand 1 താപനിയന്ത്രണം definition/meaning താപനിയന്ത്രണം. താപനിയന്ത്രണം താപനിയന്ത്രണം, steps, application താപനിയന്ത്രണം താപനിയന്ത്രണം. Technical terms English-ൽ താപനിയന്ത്രണം താപനിയന്ത്രണം.

## ▼ Systems prevent unsafe increases in pressure or temperature.

Systems prevent unsafe increases in pressure or temperature. is an official topic in Module 3 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Systems prevent unsafe increases in pressure or temperature. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, systems prevent unsafe increases in pressure or temperature. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect | What to prepare |
|--------|-----------------|
|--------|-----------------|

| Aspect              | What to prepare                                                                                        |
|---------------------|--------------------------------------------------------------------------------------------------------|
| Definition/identity | What Systems prevent unsafe increases in pressure or temperature. means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                      |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-□□ Systems prevent unsafe increases in pressure or temperature. □□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□□, steps, application □□□□□□ □□□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

## ▼ Summarise fire and explosion prevention systems.

Summarise fire and explosion prevention systems. is an official topic in Module 3 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Summarise fire and explosion prevention systems. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, summarise fire and explosion prevention systems. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                            |
|---------------------|--------------------------------------------------------------------------------------------|
| Definition/identity | What Summarise fire and explosion prevention systems. means in the official course context |



## ▼ Fire and Explosion Recognise the basic function of flame arresters' inerting

Fire and Explosion Recognise the basic function of flame arresters' inerting is an official topic in Module 3 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Fire and Explosion Recognise the basic function of flame arresters' inerting using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, fire and explosion recognise the basic function of flame arresters' inerting should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect | What to prepare |
|--------|-----------------|
|--------|-----------------|



| Aspect              | What to prepare                                                                                                        |
|---------------------|------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Fire and Explosion Recognise the basic function of flame arresters' inerting means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                      |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-☐☐ Fire and Explosion Recognise the basic function of flame arresters' inerting ☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐ definition/meaning ☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐. ☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐, steps, application ☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐. Technical terms English-☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐.

## ▼ Prevention Systems and blanketing systems in preventing ignition or fire

Prevention Systems and blanketing systems in preventing ignition or fire is an official topic in Module 3 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Prevention Systems and blanketing systems in preventing ignition or fire using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, prevention systems and blanketing systems in preventing ignition or fire should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect | What to prepare |
|--------|-----------------|
|--------|-----------------|

**Malayalam support:** Module 3-  
Prevention Systems and blanketing systems in preventing ignition or fire  
definition/meaning .  
application .  
Technical terms English-  
.

## ▼ spread.

spread. is an official topic in Module 3 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify spread. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, spread. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                         |
|---------------------|---------------------------------------------------------|
| Definition/identity | What spread. means in the official course context       |
| Study lens          | Principle → components or stages → result → application |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-എന്നു വിതരിച്ചിരിക്കുന്നു. നിർവ്വചനം/അർത്ഥം നിർവ്വചിക്കുക. നിർവ്വചനം നിർവ്വചിക്കുക, steps, application നിർവ്വചിക്കുക നിർവ്വചിക്കുക. Technical terms English-ൽ നിർവ്വചിക്കുക.

## ▼ Summarise common types of detectors (gas detectors,

Summarise common types of detectors (gas detectors, is an official topic in Module 3 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Summarise common types of detectors (gas detectors, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, summarise common types of detectors (gas detectors, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                               |
|---------------------|-----------------------------------------------------------------------------------------------|
| Definition/identity | What Summarise common types of detectors (gas detectors, means in the official course context |



## ▼ Hazard Detection and

Hazard Detection and is an official topic in Module 3 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Hazard Detection and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, hazard detection and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                |
|---------------------|----------------------------------------------------------------|
| Definition/identity | What Hazard Detection and means in the official course context |
| Study lens          | Principle → components or stages → result → application        |





## ▼ flame and smoke detectors) and state their warning L CO3 Understand 1

flame and smoke detectors) and state their warning L CO3 Understand 1 is an official topic in Module 3 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify flame and smoke detectors) and state their warning L CO3 Understand 1 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, flame and smoke detectors) and state their warning L CO3 understand 1 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect | What to prepare |
|--------|-----------------|
|--------|-----------------|

| Aspect              | What to prepare                                                                                                 |
|---------------------|-----------------------------------------------------------------------------------------------------------------|
| Definition/identity | What flame and smoke detectors) and state their warning L CO3 Understand 1 means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                         |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                               |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3- flame and smoke detectors) and state their warning L CO3 Understand 1 definition/meaning . . . . . definition/meaning . . . . . steps, application . . . . . Technical terms English- . . . . .

## ▼ Monitoring Systems

Monitoring Systems is an official topic in Module 3 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Monitoring Systems using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, monitoring systems should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                              |
|---------------------|--------------------------------------------------------------|
| Definition/identity | What Monitoring Systems means in the official course context |
| Study lens          | Principle → components or stages → result → application      |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-□□ Monitoring Systems □□□□□□□□□□  
□□□□□ definition/meaning □□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□□, steps,  
 application □□□□□□ □□□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□□  
□□□□□□□□□□□□.

► **Official syllabus detail and terminology (expand in digital version)**

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

**MODULE 4**

# Fire extinguishing Explain fire-extinguishing techniques - cooling,

This module is presented from the official syllabus scope for Industrial Safety - Principles and Practices. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**  
official hours

**CO-linked**  
see outcomes

## Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

## Key facts

- Official module title: Fire extinguishing Explain fire-extinguishing techniques - cooling,
- Official duration: As specified
- Official topic entries captured: 24
- The full source excerpt is preserved below for coverage checking.

## ▼ Fire extinguishing Explain fire-extinguishing techniques - cooling,

Fire extinguishing Explain fire-extinguishing techniques - cooling, is an official topic in Module 4 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Fire extinguishing Explain fire-extinguishing techniques - cooling, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, fire extinguishing explain fire-extinguishing techniques - cooling, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect | What to prepare |
|--------|-----------------|
|--------|-----------------|





## ▼ techniques smothering, starving, chain breaking

techniques smothering, starving, chain breaking is an official topic in Module 4 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify techniques smothering, starving, chain breaking using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, techniques smothering, starving, chain breaking should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                           |
|---------------------|-------------------------------------------------------------------------------------------|
| Definition/identity | What techniques smothering, starving, chain breaking means in the official course context |



## ▼ Fire extinguishing Classify fires based on the type of combustible material

Fire extinguishing Classify fires based on the type of combustible material is an official topic in Module 4 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Fire extinguishing Classify fires based on the type of combustible material using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, fire extinguishing classify fires based on the type of combustible material should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect | What to prepare |
|--------|-----------------|
|--------|-----------------|



## ▼ system classification and relate them to suitable extinguishing media.

system classification and relate them to suitable extinguishing media. is an official topic in Module 4 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify system classification and relate them to suitable extinguishing media. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, system classification and relate them to suitable extinguishing media. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect | What to prepare |
|--------|-----------------|
|--------|-----------------|

| Aspect              | What to prepare                                                                                                  |
|---------------------|------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What system classification and relate them to suitable extinguishing media. means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                          |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-എന്ന സിസ്റ്റം ക്ലാസിഫിക്കേഷൻ ആയിട്ടുള്ളതും അതിനെ തക്കതായ അഗ്നിശമന മാധ്യമം (suitable extinguishing media) ന്നതും വിശദീകരിക്കുക. നിർവ്വചനം/അർത്ഥം (definition/meaning) വിശദീകരിക്കുക. പ്രിൻസിപ്പിൾ (principle), ഘട്ടങ്ങൾ (steps), പ്രയോഗം (application) വിശദീകരിക്കുക. Technical terms English-ൽ എഴുതുക. Technical terms English-ൽ എഴുതുക.

## ▼ Water-type Describe the general operating principle, suitability, and

Water-type Describe the general operating principle, suitability, and is an official topic in Module 4 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Water-type Describe the general operating principle, suitability, and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, water-type describe the general operating principle, suitability, and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect | What to prepare |
|--------|-----------------|
|--------|-----------------|

| Aspect              | What to prepare                                                                                                 |
|---------------------|-----------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Water-type Describe the general operating principle, suitability, and means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                         |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                               |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4- Water-type Describe the general operating principle, suitability, and definition/meaning. steps, application. Technical terms English-.



## ▼ Extinguishers limitations of water-type extinguishers.

Extinguishers limitations of water-type extinguishers. is an official topic in Module 4 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Extinguishers limitations of water-type extinguishers. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, extinguishers limitations of water-type extinguishers. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                                  |
|---------------------|--------------------------------------------------------------------------------------------------|
| Definition/identity | What Extinguishers limitations of water-type extinguishers. means in the official course context |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4- Extinguishers limitations of water-type extinguishers. definition/meaning .  
 , steps, application Technical terms English-

## ▼ Explain the basic principle of foam extinguishers and

Explain the basic principle of foam extinguishers and is an official topic in Module 4 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Explain the basic principle of foam extinguishers and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, explain the basic principle of foam extinguishers and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                                 |
|---------------------|-------------------------------------------------------------------------------------------------|
| Definition/identity | What Explain the basic principle of foam extinguishers and means in the official course context |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Study lens    | Principle → components or stages → result → application                           |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-എം Explain the basic principle of foam extinguishers and എം definition/meaning എം. എം steps, application എം. Technical terms English-എം എം.

## ▼ Foam-type extinguishers identify situations where they are used and their L CO4 Understand 1

Foam-type extinguishers identify situations where they are used and their L CO4 Understand 1 is an official topic in Module 4 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Foam-type extinguishers identify situations where they are used and their L CO4 Understand 1 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, foam-type extinguishers identify situations where they are used and their L CO4 Understand 1 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                                                                        |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Foam-type extinguishers identify situations where they are used and their L CO4 Understand 1 means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                                |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                      |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-ഫോം ടൈപ്പ് എക്സ്റ്റിംഗ്വിഷറുകൾ ഐഡൻ്റൈസ് ചെയ്യുന്ന സാഹചര്യങ്ങൾ അവയുടെ L CO4 Understand 1 ന്റെ അർത്ഥം വ്യക്തമാക്കുക. definition/meaning വ്യക്തമാക്കുക. ഐഡൻ്റൈസ് ചെയ്യുന്ന സാഹചര്യങ്ങൾ, steps, application വ്യക്തമാക്കുക. Technical terms English-ൽ വ്യക്തമാക്കുക.

## ▼ limitations.

limitations. is an official topic in Module 4 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify limitations. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, limitations. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                         |
|---------------------|---------------------------------------------------------|
| Definition/identity | What limitations. means in the official course context  |
| Study lens          | Principle → components or stages → result → application |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4- $\square$  limitations.  $\square$  definition/meaning  $\square$ .  $\square$  steps, application  $\square$ . Technical terms English- $\square$   $\square$ .



## ▼ Explain how dry chemical powder (DCP) extinguishers

Explain how dry chemical powder (DCP) extinguishers is an official topic in Module 4 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Explain how dry chemical powder (DCP) extinguishers using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, explain how dry chemical powder (dcp) extinguishers should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                               |
|---------------------|-----------------------------------------------------------------------------------------------|
| Definition/identity | What Explain how dry chemical powder (DCP) extinguishers means in the official course context |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Study lens    | Principle → components or stages → result → application                           |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-എന്നു വിശദീകരിക്കുക (Explain how dry chemical powder (DCP) extinguishers work) എന്നതിന് നിർവ്വചനം/അർത്ഥം എഴുതുക. എന്നതിന് നിർവ്വചനം, steps, application എന്നിവയെക്കുറിച്ചുള്ള വിവരങ്ങൾ. Technical terms English-ൽ എഴുതുക.

## ▼ Dry Chemical Powder

Dry Chemical Powder is an official topic in Module 4 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Dry Chemical Powder using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, dry chemical powder should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                               |
|---------------------|---------------------------------------------------------------|
| Definition/identity | What Dry Chemical Powder means in the official course context |
| Study lens          | Principle → components or stages → result → application       |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-ഓക്സിജൻ Dry Chemical Powder ഓക്സിജൻ ഓക്സിജൻ definition/meaning ഓക്സിജൻ ഓക്സിജൻ. ഓക്സിജൻ ഓക്സിജൻ ഓക്സിജൻ, steps, application ഓക്സിജൻ ഓക്സിജൻ ഓക്സിജൻ. Technical terms English-ഓക്സിജൻ ഓക്സിജൻ.

## ▼ interrupt combustion. State their applications and L CO4 Understand 1

interrupt combustion. State their applications and L CO4 Understand 1 is an official topic in Module 4 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify interrupt combustion. State their applications and L CO4 Understand 1 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, interrupt combustion. state their applications and I co4 understand 1 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect | What to prepare |
|--------|-----------------|
|--------|-----------------|

| Aspect              | What to prepare                                                                                                 |
|---------------------|-----------------------------------------------------------------------------------------------------------------|
| Definition/identity | What interrupt combustion. State their applications and L CO4 Understand 1 means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                         |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                               |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-ഇന്റർപ്രീറ്റ് കമ്പ്യൂട്ടേഷൻ. അവയുടെ പ്രയോഗങ്ങൾ ഉൾക്കൊള്ളിച്ച് L CO4 Understand 1 ഇന്റർപ്രീറ്റ് കമ്പ്യൂട്ടേഷൻ എന്നതിന്റെ നിർവ്വചനം/അർത്ഥം എഴുതുക. ഇന്റർപ്രീറ്റ് കമ്പ്യൂട്ടേഷൻ, steps, application എന്നിവയെക്കുറിച്ച് വിശദീകരിക്കുക. Technical terms English-ൽ എഴുതുക. ഇന്റർപ്രീറ്റ് കമ്പ്യൂട്ടേഷൻ.

## ▼ (DCP) Extinguishers

(DCP) Extinguishers is an official topic in Module 4 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify (DCP) Extinguishers using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, (dcp) extinguishers should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                               |
|---------------------|---------------------------------------------------------------|
| Definition/identity | What (DCP) Extinguishers means in the official course context |
| Study lens          | Principle → components or stages → result → application       |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-പ്രതിരോധ (DCP) Extinguishers പദപരിചയം  
പദപരിചയം definition/meaning പദപരിചയം പദപരിചയം. പദപരിചയം പദപരിചയം പദപരിചയം, steps,  
application പദപരിചയം പദപരിചയം പദപരിചയം. Technical terms English-ൽ പദപരിചയം  
പദപരിചയം പദപരിചയം.



## ▼ Carbon Dioxide (CO<sub>2</sub>) Explain the smothering action of CO<sub>2</sub> extinguishers,

Carbon Dioxide (CO<sub>2</sub>) Explain the smothering action of CO<sub>2</sub> extinguishers, is an official topic in Module 4 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Carbon Dioxide (CO<sub>2</sub>) Explain the smothering action of CO<sub>2</sub> extinguishers, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, carbon dioxide (co<sub>2</sub>) explain the smothering action of co<sub>2</sub> extinguishers, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect | What to prepare |
|--------|-----------------|
|--------|-----------------|

| Aspect              | What to prepare                                                                                                                             |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Carbon Dioxide (CO <sub>2</sub> ) Explain the smothering action of CO <sub>2</sub> extinguishers, means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                                     |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                           |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-കാർബൺ ഡയോക്സൈഡ് (CO<sub>2</sub>) Explain the smothering action of CO<sub>2</sub> extinguishers, നിർവ്വചനം/അർത്ഥം, പ്രയോഗം, definition/meaning, steps, application. Technical terms English-ൽ ഉൾപ്പെടെ.

## ▼ Fire Extinguishers typical uses and limitations.

Fire Extinguishers typical uses and limitations. is an official topic in Module 4 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Fire Extinguishers typical uses and limitations. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, fire extinguishers typical uses and limitations. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                            |
|---------------------|--------------------------------------------------------------------------------------------|
| Definition/identity | What Fire Extinguishers typical uses and limitations. means in the official course context |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Study lens    | Principle → components or stages → result → application                           |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-🔥 Fire Extinguishers typical uses and limitations. 📄📄📄📄📄📄📄📄📄📄📄 definition/meaning 📄📄📄📄📄📄📄📄. 📄📄📄📄📄📄📄📄📄📄📄, steps, application 📄📄📄📄📄📄📄📄📄📄📄. Technical terms English-📄 📄📄📄📄📄📄📄📄📄📄📄.

## ▼ Recognise special extinguishers used for metal fires,

Recognise special extinguishers used for metal fires, is an official topic in Module 4 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Recognise special extinguishers used for metal fires, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, recognise special extinguishers used for metal fires, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                                 |
|---------------------|-------------------------------------------------------------------------------------------------|
| Definition/identity | What Recognise special extinguishers used for metal fires, means in the official course context |



## ▼ Special Extinguishers L CO4 Understand 1

Special Extinguishers L CO4 Understand 1 is an official topic in Module 4 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Special Extinguishers L CO4 Understand 1 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, special extinguishers I co4 understand 1 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                    |
|---------------------|------------------------------------------------------------------------------------|
| Definition/identity | What Special Extinguishers L CO4 Understand 1 means in the official course context |





## ▼ their purpose and limitations.

their purpose and limitations. is an official topic in Module 4 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify their purpose and limitations. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, their purpose and limitations. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                          |
|---------------------|--------------------------------------------------------------------------|
| Definition/identity | What their purpose and limitations. means in the official course context |
| Study lens          | Principle → components or stages → result → application                  |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-ഇരുപതു തിരക്കുകളുടെ ഉദ്ദേശ്യം പരിമിതികൾ.

ഇരുപതു തിരക്കുകളുടെ definition/meaning ഇരുപതു തിരക്കുകളുടെ. ഇരുപതു തിരക്കുകളുടെ steps, application ഇരുപതു തിരക്കുകളുടെ. Technical terms English-ഇരുപതു തിരക്കുകളുടെ.

## ▼ Explain why PPE is necessary and recognise its

Explain why PPE is necessary and recognise its is an official topic in Module 4 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Explain why PPE is necessary and recognise its using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, explain why ppe is necessary and recognise its should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                          |
|---------------------|------------------------------------------------------------------------------------------|
| Definition/identity | What Explain why PPE is necessary and recognise its means in the official course context |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Study lens    | Principle → components or stages → result → application                           |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-എന്നു പറയുന്ന PPE ആവശ്യമാണ് എന്ന് വിശദീകരിക്കുകയും അതിന്റെ നിർവ്വചനം/അർത്ഥം എന്താണെന്ന് മനസ്സിലാക്കുകയും ചെയ്യുക. നിർവ്വചനം, ഘട്ടങ്ങൾ, പ്രയോഗം എന്നിവയെക്കുറിച്ചും വിശദീകരിക്കുകയും ചെയ്യുക. Technical terms English-ൽ എഴുതുകയും വിശദീകരിക്കുകയും ചെയ്യുക.

## ▼ Requirements of PPE L CO4 Understand 1

Requirements of PPE L CO4 Understand 1 is an official topic in Module 4 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Requirements of PPE L CO4 Understand 1 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, requirements of ppe l co4 understand 1 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                  |
|---------------------|----------------------------------------------------------------------------------|
| Definition/identity | What Requirements of PPE L CO4 Understand 1 means in the official course context |



## ▼ limitations as the last line of defence.

limitations as the last line of defence. is an official topic in Module 4 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify limitations as the last line of defence. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, limitations as the last line of defence. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                    |
|---------------------|------------------------------------------------------------------------------------|
| Definition/identity | What limitations as the last line of defence. means in the official course context |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Study lens    | Principle → components or stages → result → application                           |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-  
 definition/meaning .  
 , steps, application . Technical terms  
 English- .



## ▼ Explain the classification of personal protective

Explain the classification of personal protective is an official topic in Module 4 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Explain the classification of personal protective using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, explain the classification of personal protective should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                             |
|---------------------|---------------------------------------------------------------------------------------------|
| Definition/identity | What Explain the classification of personal protective means in the official course context |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Study lens    | Principle → components or stages → result → application                           |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4- Explain the classification of personal protective equipment definition/meaning. Explain steps, application. Technical terms English- Malayalam.

## ▼ Types of PPE equipment: types of PPE, head protection, eye and face L CO4 Understand 2

Types of PPE equipment: types of PPE, head protection, eye and face L CO4 Understand 2 is an official topic in Module 4 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Types of PPE equipment: types of PPE, head protection, eye and face L CO4 Understand 2 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, types of ppe equipment: types of ppe, head protection, eye and face l co4 understand 2 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect | What to prepare |
|--------|-----------------|
|--------|-----------------|



## ▼ protection, Respiratory protection, and Body protection

protection, Respiratory protection, and Body protection is an official topic in Module 4 of Industrial Safety - Principles and Practices. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify protection, Respiratory protection, and Body protection using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, protection, respiratory protection, and body protection should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                                   |
|---------------------|---------------------------------------------------------------------------------------------------|
| Definition/identity | What protection, Respiratory protection, and Body protection means in the official course context |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Study lens    | Principle → components or stages → result → application                           |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-പ്രതിരോധം, ശ്വാസരോധം, and Body protection പദങ്ങളുടെ definition/meaning പദങ്ങളുടെ. പദങ്ങളുടെ പദങ്ങളുടെ, steps, application പദങ്ങളുടെ പദങ്ങളുടെ. Technical terms English-ൽ പദങ്ങളുടെ പദങ്ങളുടെ.

### ► Official syllabus detail and terminology (expand in digital version)

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

**RAPID REFERENCE****Concept and formula bank****Answer structure**

Definition → Principle →  
Components/steps →  
Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

**Practical record**

Aim → Apparatus → Procedure  
→ Observation → Result →  
Safety

Use this sequence for laboratory, workshop and field activities.

**RAPID REVISION**

## Diagram and source-transcript library

### **Module 1 source and topic cards**

Jump to official terminology, topic explanations and study guidance.

### **Module 2 source and topic cards**

Jump to official terminology, topic explanations and study guidance.

### **Module 3 source and topic cards**

Jump to official terminology, topic explanations and study guidance.

### **Module 4 source and topic cards**

Jump to official terminology, topic explanations and study guidance.



**OFFICIAL SELF-LEARNING**

## Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

- Module IV
- Fire extinguishing Explain fire-extinguishing techniques - cooling, L CO4 Understand 2
- techniques smothering, starving, chain breaking
- Fire extinguishing Classify fires based on the type of combustible material L CO4 Understand 2
- system classification and relate them to suitable extinguishing media.
- Water-type Describe the general operating principle, suitability, and L CO4 Understand 2
- Extinguishers limitations of water-type extinguishers.
- Explain the basic principle of foam extinguishers and
- Foam-type extinguishers identify situations where they are used and their L CO4 Understand 1
- limitations.
- Explain how dry chemical powder (DCP) extinguishers Dry Chemical Powder
- interrupt combustion. State their applications and L CO4 Understand 1 (DCP) Extinguishers
- Carbon Dioxide (CO<sub>2</sub>) Explain the smothering action of CO<sub>2</sub> extinguishers, L CO4 Understand 1
- Fire Extinguishers typical uses and limitations.
- Recognise special extinguishers used for metal fires,
- Special Extinguishers L CO4 Understand 1
- their purpose and limitations.
- Explain why PPE is necessary and recognise its
- Requirements of PPE L CO4 Understand 1
- limitations as the last line of defence.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

|               |               |               |               |            |     |
|---------------|---------------|---------------|---------------|------------|-----|
| – Theory      |               |               |               |            |     |
| CA3           | CA4           | CA5           | CA1           | CA6        | CA2 |
| ESE           |               |               |               |            |     |
| Mode          | Attendance    | Self-learning | Self-learning |            |     |
| Self-learning | Self-learning | Series        | Series        |            |     |
| Written       |               |               |               |            |     |
| activities    | activities    | activities    |               | activities |     |
| Exam          |               | Exam          | Exam          |            |     |
| (Module 3)    | (Module 4)    | (Module 1)    | (Module 2)    |            |     |
| (theory)      |               | (2 modules)   | (2 modules)   |            |     |
| Duration      |               |               |               |            |     |
| 2 hours       | 2 hours       | 2.5 hours     |               |            |     |
| Exam mark     |               |               | 15            |            | 15  |
| 15            | 15            | 50            |               | 50         |     |
| 60            |               |               |               |            |     |
| Converted to  |               |               |               |            |     |
| 20            | 20            |               |               |            |     |
| Marks         |               | 5             |               |            |     |
| 15            |               |               |               | 20         |     |
| 60            |               |               |               |            |     |
| Total         |               |               |               |            |     |
| 40            |               |               |               |            |     |
| 60            |               |               |               |            |     |

**Practice-paper notice:** The model paper below is generated for revision and is not an official examination paper.

**ACTIVE RECALL**

## Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

### Module 1

► **Define or explain Industrial Safety temperature - toxicity - routes of entry - threshold exposure limits - chemistry of fire -. (3 marks)**

► **Define or explain fire triangle - fire tetrahedron - classification of fires - causes of fire - major industrial. (3 marks)**

► **Define or explain accidents - safety lessons.. (3 marks)**

► **Define or explain Hierarchy of controls - significance in risk reduction - source-level control - inherently. (3 marks)**

### Module 2

► **Define or explain Inherently Safer safety benefits - inventory reduction - safer process routes - advantages and 15 0. (3 marks)**

► **Define or explain Approaches limitations of source-level control - awareness of hazard analysis methods (PHA -. (3 marks)**

► **Define or explain What-If - JSA - HAZOP - FTA) - purpose and basic application of each method.. (3 marks)**

► **Define or explain Engineering controls - limitations of engineering controls - process containment -. (3 marks)**

## Module 3

► Define or explain Define engineering control and explain its purpose.. (3 marks)

► Define or explain Introduction to Differentiate between source control and engineering. (3 marks)

► Define or explain Engineering Controls control.. (3 marks)

► Define or explain Identify the limitations of engineering control.. (3 marks)

## Module 4

► Define or explain Fire extinguishing Explain fire-extinguishing techniques - cooling,. (3 marks)

► Define or explain techniques smothering, starving, chain breaking. (3 marks)

► Define or explain Fire extinguishing Classify fires based on the type of combustible material. (3 marks)

► Define or explain system classification and relate them to suitable extinguishing media.. (3 marks)

**PRACTICE ONLY**

## Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

### Module 1

1-mark definition: State the meaning of **Industrial Safety temperature - toxicity - routes of entry - threshold exposure limits - chemistry of fire -**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

### Module 2

1-mark definition: State the meaning of **Inherently Safer safety benefits - inventory reduction - safer process routes - advantages and 15 0.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

### Module 3

1-mark definition: State the meaning of **Define engineering control and explain its purpose..**

3-mark explanation: Explain one principle, process or comparison from

### Module 4

1-mark definition: State the meaning of **Fire extinguishing Explain fire-extinguishing techniques - cooling,.**

this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

## QUICK REVISION

### Exam checklist

#### Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

#### Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

**REFERENCE VOCABULARY**

# Glossary

**Study glossary**

| Term              | Meaning in this lesson                                                   |
|-------------------|--------------------------------------------------------------------------|
| Course Outcome    | A capability the student should demonstrate after completing the course. |
| Module            | An official syllabus unit with defined topics and hours.                 |
| CIE               | Continuous Internal Evaluation.                                          |
| ESE               | End Semester Examination.                                                |
| Source transcript | A preserved extract of the official syllabus used for coverage checking. |

**SOURCE-SUPPORTED REFERENCES**

## References and web resources

**References listed in the official PDF**

References are included in the official source PDF.

**Web resources listed in the official PDF**

- URL Modules Covered
- <https://www.nptelprep.in/courses/110105094> I,II,III
- [https://onlinecourses.swayam2.ac.in/nou26\\_ge33/preview](https://onlinecourses.swayam2.ac.in/nou26_ge33/preview) I,II,IV

**IN-PAGE SEARCH****Search this lesson**

Prepared from the supplied official SITTTTR syllabus PDF for Course 2073. Verify institutional updates against the current official curriculum.